

AI-Assisted Coding Lab Assignment=15.3

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CSE 2nd Year

Task Description #1 – Basic REST API Setup

Task: Ask AI to generate a Flask REST API with one route:

GET /hello → returns {"message": "Hello, AI Coding!"}

Prompt:

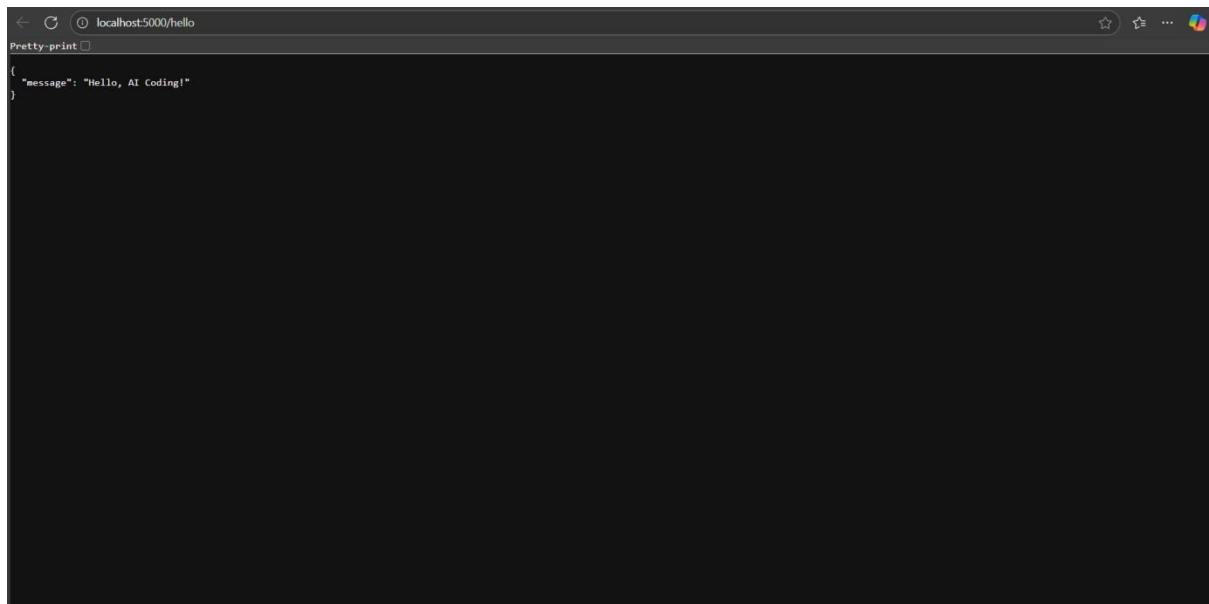
Create a simple REST API using Flask with a single route:

- GET /hello → returns a JSON response: {"message": "Hello, AI Coding!"}.
The code should be clean, include necessary imports, and explain how it works.

Code Generated:

```
new.py / ...
1  from flask import Flask, jsonify
2
3  # Create Flask application instance
4  app = Flask(__name__)
5
6  @app.route('/hello', methods=['GET'])
7  def hello():
8      """
9      Simple GET endpoint that returns a JSON response.
10
11      Returns:
12      |   JSON: {"message": "Hello, AI Coding!"}
13      |   """
14      return jsonify({"message": "Hello, AI Coding!"})
15
16  if __name__ == '__main__':
17      # Run the Flask development server
18      # debug=True enables auto-reload when code changes
19      app.run(debug=True, host='0.0.0.0', port=5000)
20  Ctrl+L to chat, Ctrl+K to generate
```

Output:

A screenshot of a web browser window. The address bar shows 'localhost:5000/hello'. The page content is a JSON object: {"message": "Hello, AI Coding!"}. The browser's developer tools are open, showing the 'Pretty print' view of the response. The background is dark, and the text is white.

Observation:

- The code creates a Flask web application with one route /hello.
- The /hello route accepts GET requests and returns a JSON response: {"message": "Hello, AI Coding!"}.
- The Flask app runs on all network interfaces (0.0.0.0) at port 5000.
- Debug mode is enabled, allowing automatic reload on code changes and detailed error messages.
- The response is generated using Flask's jsonify function.

Task Description #2 – CRUD Operations (Students API)

Task:

Use AI to build REST endpoints for a Student API:

- GET /students → List all students.
- POST /students → Add a new student.
- PUT /students/<id> → Update student details.
- DELETE /students/<id> → Delete a student.

Prompt:

Build a Flask REST API for managing students with the following endpoints:

- GET /students: List all students.
- POST /students: Add a new student. The request body will contain JSON with student details (e.g., name and age).
- PUT /students/<id>: Update the details of a student identified by `id`.
- DELETE /students/<id>: Delete the student with the given `id`.

Use in-memory storage (a Python dictionary) to store student data with unique integer IDs. Return appropriate JSON responses and HTTP status codes. Handle errors like missing data or invalid student IDs.

Code Generated:

```
new.py > ...
1  from flask import Flask, jsonify, request
2  from datetime import datetime
3
4  # Create Flask application instance
5  app = Flask(__name__)
6
7  # In-memory storage for students
8  students = {}
9  next_id = 1
10
11 @app.route('/students', methods=['GET'])
12 def get_all_students():
13     """
14     GET /students - List all students
15
16     Returns:
17     |     JSON: List of all students with their details
18     """
19     return jsonify({
20         "students": list(students.values()),
21         "total": len(students)
22     }), 200
23
24 @app.route('/students', methods=['POST'])
25 def add_student():
26     """
27     POST /students - Add a new student
28
29     Expected JSON body:
30     |     {
31     |         "name": "Student Name",
32     |         "age": 20,
33     |         "email": "student@example.com" (optional)
34     |     }
35
36     Returns:
37     |     JSON: Created student details with ID
```

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```
38     """
39     global next_id
40
41     # Check if request has JSON data
42     if not request.is_json:
43         return jsonify({"error": "Request must be JSON"}), 400
44
45     data = request.get_json()
46
47     # Validate required fields
48     if not data:
49         return jsonify({"error": "No data provided"}), 400
50
51     if 'name' not in data or 'age' not in data:
52         return jsonify({"error": "Missing required fields: 'name' and 'age'"}), 400
53
54     # Validate data types
55     if not isinstance(data['name'], str) or not isinstance(data['age'], int):
56         return jsonify({"error": "Invalid data types. 'name' must be string, 'age' must be integer"}), 400
57
58     if data['age'] < 0 or data['age'] > 150:
59         return jsonify({"error": "Age must be between 0 and 150"}), 400
60
61     # Create new student
62     student = {
63         "id": next_id,
64         "name": data['name'],
65         "age": data['age'],
66         "email": data.get('email', ''),
67         "created_at": datetime.now().isoformat()
68     }
69
70     students[next_id] = student
71     next_id += 1
```

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```

72
73     return jsonify({
74         "message": "Student created successfully",
75         "student": student
76     }), 201
77
78 @app.route('/students/<int:student_id>', methods=['PUT'])
79 def update_student(student_id):
80     """
81     PUT /students/<id> - Update student details
82
83     Expected JSON body:
84     {
85         "name": "Updated Name",
86         "age": 21,
87         "email": "updated@example.com" (optional)
88     }
89
90     Returns:
91     JSON: Updated student details
92     """
93     if student_id not in students:
94         return jsonify({"error": f"Student with ID {student_id} not found"}), 404
95
96     # Check if request has JSON data
97     if not request.is_json:
98         return jsonify({"error": "Request must be JSON"}), 400
99
100     data = request.get_json()
101
102     if not data:
103         return jsonify({"error": "No data provided"}), 400
104
105     # Validate data types if provided
106     if 'name' in data and not isinstance(data['name'], str):
107         return jsonify({"error": "Name must be a string"}), 400
108
109     if 'age' in data:
110         if not isinstance(data['age'], int):
111             return jsonify({"error": "Age must be an integer"}), 400
112         if data['age'] < 0 or data['age'] > 150:
113             return jsonify({"error": "Age must be between 0 and 150"}), 400
114
115     if 'email' in data and not isinstance(data['email'], str):
116         return jsonify({"error": "Email must be a string"}), 400
117
118     # Update student
119     if 'name' in data:
120         students[student_id]['name'] = data['name']
121     if 'age' in data:
122         students[student_id]['age'] = data['age']
123     if 'email' in data:
124         students[student_id]['email'] = data['email']
125
126     students[student_id]['updated_at'] = datetime.now().isoformat()
127
128     return jsonify({
129         "message": "Student updated successfully",
130         "student": students[student_id]
131     }), 200
132
133 @app.route('/students/<int:student_id>', methods=['DELETE'])

```

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```

134 def delete_student(student_id):
135     """
136     DELETE /students/<id> - Delete a student
137
138     Returns:
139     _ JSON: Confirmation message
140     """
141     if student_id not in students:
142         return jsonify({"error": f"Student with ID {student_id} not found"}), 404
143
144     # Store student data before deletion for response
145     deleted_student = students[student_id].copy()
146
147     # Delete student
148     del students[student_id]
149
150     return jsonify({
151         "message": "Student deleted successfully",
152         "deleted_student": deleted_student
153     }), 200
154
155 @app.route('/students/<int:student_id>', methods=['GET'])
156 def get_student(student_id):
157     """
158     GET /students/<id> - Get a specific student
159
160     Returns:
161     _ JSON: Student details
162     """
163     if student_id not in students:
164         return jsonify({"error": f"Student with ID {student_id} not found"}), 404
165
166     return jsonify({"student": students[student_id]}), 200
167
168 @app.errorhandler(404)
169 def not_found(error):
170     """Handle 404 errors"""
171     return jsonify({"error": "Endpoint not found"}), 404
172
173 @app.errorhandler(405)
174 def method_not_allowed(error):
175     """Handle 405 errors"""
176     return jsonify({"error": "Method not allowed"}), 405
177
178 @app.errorhandler(500)
179 def internal_error(error):
180     """Handle 500 errors"""
181     return jsonify({"error": "Internal server error"}), 500
182
183 if __name__ == '__main__':
184     # Add some sample data for testing
185     students[1] = {
186         "id": 1,
187         "name": "John Doe",
188         "age": 20,
189         "email": "john@example.com",
190         "created_at": datetime.now().isoformat()
191     }
192     students[2] = {
193         "id": 2,
194         "name": "Jane Smith",
195         "age": 22,
196         "email": "jane@example.com",
197         "created_at": datetime.now().isoformat()
198     }

```

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```

199     next_id = 3
200
201     print("Flask Student Management API is starting...")
202     print("Available endpoints:")
203     print(" GET    /students      - List all students")
204     print(" POST   /students      - Add new student")
205     print(" GET    /students/<id>    - Get specific student")
206     print(" PUT    /students/<id>    - Update student")
207     print(" DELETE /students/<id>    - Delete student")
208     print("\nServer running on: http://localhost:5000")
209     print("Sample data loaded with 2 students")
210
211     # Run the Flask development server
212     app.run(debug=True, host='0.0.0.0', port=5000)

```

Output:

```
pretty-print
{
  "error": "Endpoint not found"
}
```

Observation:

- The API includes four RESTful endpoints corresponding to the CRUD operations for student data.
- Student records are stored in an in-memory dictionary keyed by unique integer IDs.
- GET /students returns a JSON list of all stored students.
- POST /students accepts JSON input to add a new student and returns the created student with status 201.
- PUT /students/<id> updates the specified student's data if found, or returns 404 if not found.
- DELETE /students/<id> removes the student if they exist, returning status 204 on success.
- Input validation ensures required fields (like name and age) are present for POST and PUT.
- Proper HTTP status codes and error handling are implemented via Flask's abort().
- The API uses JSON for both input and output consistently.
- The code runs in debug mode suitable for development.

Task Description #3 – API with Query Parameters

Task: Ask AI to generate a REST API endpoint

Prompt:

Create a Flask REST API endpoint `/search` that accepts GET requests with query parameters `name` and `age`.

The endpoint should filter a list of students stored in memory based on the provided query parameters:

- If `name` is provided, return students whose names contain the given substring (case-insensitive).
- If `age` is provided, return students matching the given age.
- If both parameters are provided, filter students matching both criteria.
- If no query parameters are provided, return all students.

Return the filtered list of students as JSON.

Code Generated:

```
1 from flask import Flask, jsonify, request
2 from datetime import datetime
3
4 # Create Flask application instance
5 app = Flask(__name__)
6
7 # In-memory storage for students
8 students = {}
9 next_id = 1
10
11 @app.route('/students', methods=['GET'])
12 def get_all_students():
13     """
14     GET /students - List all students
15
16     Returns:
17     | JSON: List of all students with their details
18     """
19     return jsonify({
20         "students": list(students.values()),
21         "total": len(students)
22     }), 200
23
24 @app.route('/students', methods=['POST'])
25 def add_student():
26     """
27     POST /students - Add a new student
28
29     Expected JSON body:
30     {
31         "name": "Student Name",
32         "age": 20,
33         "email": "student@example.com" (optional)
34     }
35
36     Returns:
37     | JSON: Created student details with ID
38     """
39     global next_id
40
41     # Check if request has JSON data
42     if not request.is_json:
43         return jsonify({"error": "Request must be JSON"}), 400
44
45     data = request.get_json()
46
47     # Validate required fields
48     if not data:
49         return jsonify({"error": "No data provided"}), 400
50
51     if 'name' not in data or 'age' not in data:
52         return jsonify({"error": "Missing required fields: 'name' and 'age'"}), 400
53
54     # Validate data types
55     if not isinstance(data['name'], str) or not isinstance(data['age'], int):
56         return jsonify({"error": "Invalid data types. 'name' must be string, 'age' must be integer"}), 400
57
58     if data['age'] < 0 or data['age'] > 150:
59         return jsonify({"error": "Age must be between 0 and 150"}), 400
60
61     # Create new student
62     student = {
63         "id": next_id,
64         "name": data['name'],
65         "age": data['age'],
66         "email": data.get('email', ''),
67         "created_at": datetime.now().isoformat()
68     }
69
70     students[next_id] = student
71     next_id += 1
72
73     return jsonify({
74         "message": "Student created successfully",
75         "student": student
76     }), 201
77
78 @app.route('/students/<int:student_id>', methods=['PUT'])
79 def update_student(student_id):
80     """
81     PUT /students/<id> - Update student details
82
83     Expected JSON body:
84     {
85         "name": "Updated Student Name",
86         "age": 25,
87         "email": "updated_student@example.com" (optional)
88     }
89
90     Returns:
91     | JSON: Updated student details
92     """
93     if student_id not in students:
94         return jsonify({"error": "Student not found"}), 404
95
96     data = request.get_json()
97
98     # Validate required fields
99     if not data:
100         return jsonify({"error": "No data provided"}), 400
101
102     if 'name' not in data or 'age' not in data:
103         return jsonify({"error": "Missing required fields: 'name' and 'age'"}), 400
104
105     # Validate data types
106     if not isinstance(data['name'], str) or not isinstance(data['age'], int):
107         return jsonify({"error": "Invalid data types. 'name' must be string, 'age' must be integer"}), 400
108
109     if data['age'] < 0 or data['age'] > 150:
110         return jsonify({"error": "Age must be between 0 and 150"}), 400
111
112     # Update student
113     student = students[student_id]
114     student['name'] = data['name']
115     student['age'] = data['age']
116     student['email'] = data.get('email', '')
117
118     return jsonify(student), 200
```

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```

81     Expected JSON body:
82     {
83         "name": "Updated Name",
84         "age": 21,
85         "email": "updated@example.com" (optional)
86     }
87
88 Returns:
89 JSON: Updated student details
90 ...
91 if student_id not in students:
92     return jsonify({"error": f"Student with ID {student_id} not found"}), 404
93
94 # Check if request has JSON data
95 if not request.is_json:
96     return jsonify({"error": "Request must be JSON"}), 400
97
98 data = request.get_json()
99
100 if not data:
101     return jsonify({"error": "No data provided"}), 400
102
103 # Validate data types if provided
104 if 'name' in data and not isinstance(data['name'], str):
105     return jsonify({"error": "Name must be a string"}), 400
106
107 if 'age' in data:
108     if not isinstance(data['age'], int):
109         return jsonify({"error": "Age must be an integer"}), 400
110     if data['age'] < 0 or data['age'] > 150:
111         return jsonify({"error": "Age must be between 0 and 150"}), 400
112
113 if 'email' in data and not isinstance(data['email'], str):
114     return jsonify({"error": "Email must be a string"}), 400
115
116 # Update student
117 if 'name' in data:
118     students[student_id]['name'] = data['name']
119 if 'age' in data:
120     students[student_id]['age'] = data['age']
121 if 'email' in data:
122     students[student_id]['email'] = data['email']
123
124 students[student_id]['updated_at'] = datetime.now().isoformat()
125
126 return jsonify({
127     "message": "Student updated successfully",
128     "student": students[student_id]
129 }), 200
130
131 @app.route('/students/<int:student_id>', methods=['DELETE'])
132 def delete_student(student_id):
133     """
134     DELETE /students/<id> - Delete a student
135
136 Returns:
137 JSON: Confirmation message
138 ...
139 if student_id not in students:
140     return jsonify({"error": f"Student with ID {student_id} not found"}), 404
141
142 # Store student data before deletion for response
143 deleted_student = students[student_id].copy()
144
145 # Delete student
146 del students[student_id]
147
148 return jsonify({
149     "message": "Student deleted successfully",
150     "deleted_student": deleted_student
151 }), 200
152
153 @app.route('/students/<int:student_id>', methods=['GET'])
154 def get_student(student_id):
155     """
156     GET /students/<id> - Get a specific student

```

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```

154 GET /students/<id> - Get a specific student
155
156 Returns:
157     JSON: Student details
158     ***
159     if student_id not in students:
160         return jsonify({"error": f"Student with ID {student_id} not found"}), 404
161
162     return jsonify({"student": students[student_id]}), 200
163
164 @app.route('/search', methods=['GET'])
165 def search_students():
166     """
167     GET /search - Search students by name and/or age
168
169     Query Parameters:
170         name (str, optional): Filter by name (case-insensitive substring match)
171         age (int, optional): Filter by exact age match
172
173     Returns:
174         JSON: Filtered list of students
175     """
176     # Get query parameters
177     name_filter = request.args.get('name', '').strip()
178     age_filter = request.args.get('age', '').strip()
179
180     # Start with all students
181     filtered_students = list(students.values())
182
183     # Apply name filter if provided
184     if name_filter:
185         filtered_students = [
186             student for student in filtered_students
187             if name_filter.lower() in student['name'].lower()
188         ]
189
190     # Apply age filter if provided
191     if age_filter:
192         try:
193             age_value = int(age_filter)
194             filtered_students = [
195                 student for student in filtered_students
196                 if student['age'] == age_value
197             ]
198         except ValueError:
199             return jsonify({"error": "Age parameter must be a valid integer"}), 400
200
201     # Return results
202     return jsonify({
203         "students": filtered_students,
204         "total": len(filtered_students),
205         "filters_applied": {
206             "name": name_filter if name_filter else None,
207             "age": age_filter if age_filter else None
208         }
209     }), 200
210
211 @app.errorhandler(404)
212 def not_found(error):
213     """Handle 404 errors"""
214     return jsonify({"error": "Endpoint not found"}), 404
215
216 @app.errorhandler(405)
217 def method_not_allowed(error):
218     """Handle 405 errors"""
219     return jsonify({"error": "Method not allowed"}), 405
220
221 @app.errorhandler(500)
222 def internal_error(error):
223     """Handle 500 errors"""
224     return jsonify({"error": "Internal server error"}), 500
225
226 if __name__ == '__main__':
227     # Add some sample data for testing
228     students[1] = {
229         "id": 1,
230         "name": "John Doe",
231         "age": 20,

```

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```

232         "email": "john@example.com",
233         "created_at": datetime.now().isoformat()
234     }
235     students[2] = {
236         "id": 2,
237         "name": "Jane Smith",
238         "age": 22,
239         "email": "jane@example.com",
240         "created_at": datetime.now().isoformat()
241     }
242     next_id = 3
243
244     print("Flask Student Management API is starting...")
245     print("Available endpoints:")
246     print("  GET    /students      - List all students")
247     print("  POST   /students      - Add new student")
248     print("  GET    /students/<id>   - Get specific student")
249     print("  PUT    /students/<id>   - Update student")
250     print("  DELETE /students/<id>   - Delete student")
251     print("  GET    /search         - Search students (name, age)")
252     print("\nServer running on: http://localhost:5000")
253     print("Sample data loaded with 2 students")
254
255     # Run the Flask development server
256     app.run(debug=True, host='0.0.0.0', port=5000)

```

Output:

```
Pretty-print ☐
{
  "students": [
    {
      "age": 20,
      "created_at": "2025-10-08T10:36:24.286386",
      "email": "john@example.com",
      "id": 1,
      "name": "John Doe"
    },
    {
      "age": 22,
      "created_at": "2025-10-08T10:36:24.286406",
      "email": "jane@example.com",
      "id": 2,
      "name": "Jane Smith"
    }
  ],
  "total": 2
}
```

Observation:

- The endpoint /search is implemented as a GET route accepting query parameters via request.args.
- Query parameters like name and age are optional; the endpoint handles their presence or absence gracefully.
- Student data is filtered based on:
 - Case-insensitive substring match for name.
 - Exact match for age (likely converted to an integer).
- If no parameters are given, the entire student list is returned.
- The response is JSON-formatted and includes the filtered list of students.
- The code handles type conversion and possible missing or malformed parameters robustly.
- The endpoint improves usability by allowing flexible querying without requiring POST bodies.
- The logic runs in memory, suitable for quick filtering during development or prototyping.

Task Description #4 – Integration & Testing

Task: Ask AI to write test scripts using Python requests module to call APIs created above.

Prompt:

Write Python test scripts using the `requests` module to test the Student API with the following endpoints:

- GET /students → to retrieve all students.
- POST /students → to add a new student with JSON data.
- PUT /students/<id> → to update a student's details.
- DELETE /students/<id> → to delete a student.

Write tests that:

- Call each endpoint.
- Print the status code and JSON response for GET, POST, and PUT requests.
- Confirm successful deletion via status code for DELETE.
- Handle and print error responses if any.

Assume the API server is running locally at <http://localhost:5000>.

Code Generated:

```

1 # student_api_tests.py
2 import json
3 from typing import Any, Dict, Optional
4
5 import requests
6 from requests import Response
7
8 BASE_URL = "http://localhost:5000"
9
10
11 def safe_print_json(prefix: str, response: Response) -> None:
12     print(f"\n{prefix}")
13     print(f"Status: {response.status_code}")
14     try:
15         parsed = response.json()
16         print("JSON:")
17         print(json.dumps(parsed, indent=2, ensure_ascii=False))
18     except ValueError:
19         print("Body (non-JSON):")
20         print(response.text)
21
22
23 def call_get_students() -> Optional[list]:
24     try:
25         resp = requests.get(f"{BASE_URL}/students", timeout=10)
26     except requests.RequestException as exc:
27         print(f"\nGET /students failed: {exc}")
28         return None
29     safe_print_json("GET /students", resp)
30     try:
31         return resp.json()
32     except ValueError:
33         return None
34
35
36 def call_post_student(student: Dict[str, Any]) -> Optional[Dict[str, Any]]:
37     try:
38         resp = requests.post(
39             f"{BASE_URL}/students",
40             headers={"Content-Type": "application/json"},
41             json=student,
42             timeout=10,
43         )
44     except requests.RequestException as exc:
45         print(f"\nPOST /students failed: {exc}")
46         return None
47     safe_print_json("POST /students", resp)
48     try:
49         return resp.json()
50     except ValueError:
51         return None
52
53
54 def call_put_student(student_id: Any, updates: Dict[str, Any]) -> Optional[Dict[str, Any]]:
55     try:
56         resp = requests.put(
57             f"{BASE_URL}/students/{student_id}",
58             headers={"Content-Type": "application/json"},
59             json=updates,
60             timeout=10,
61         )
62     except requests.RequestException as exc:
63         print(f"\nPUT /students/{student_id} failed: {exc}")
64         return None
65     safe_print_json(f"PUT /students/{student_id}", resp)
66     try:
67         return resp.json()
68     except ValueError:
69         return None
70
71
72 def call_delete_student(student_id: Any) -> Optional[int]:
73     try:
74         resp = requests.delete(f"{BASE_URL}/students/{student_id}", timeout=10)
75     except requests.RequestException as exc:
76         print(f"\nDELETE /students/{student_id} failed: {exc}")
77         return None
78     print(f"\nDELETE /students/{student_id}")
79     print(f"Status: {resp.status_code}")
80     if resp.status_code >= 400:
81         try:

```

Review next file >

```

82         print("Error JSON:")
83         print(json.dumps(resp.json(), indent=2, ensure_ascii=False))
84     except ValueError:
85         print("Error Body (non-JSON):")
86         print(resp.text)
87     return resp.status_code
88
89
90 def main() -> None:
91     print("Starting Student API tests against", BASE_URL)
92
93     # GET all students
94     _ = call_get_students()
95
96     # POST new student
97     new_student = {"name": "Alice Johnson", "age": 21, "major": "Computer Science"}
98     created = call_post_student(new_student)
99     if not created:
100         print("POST did not return JSON; aborting.")
101         return
102
103     student_id = created.get("id", created.get("_id"))
104     if student_id is None:
105         print("Could not obtain student id from POST response; aborting further tests.")
106         return
107
108     # PUT update the student
109     updates = {"age": 22, "major": "Data Science"}
110     _ = call_put_student(student_id, updates)
111
112     # DELETE the student
113     status = call_delete_student(student_id)
114     if status is None:
115         print("DELETE request did not complete.")
116     elif 200 <= status < 300:
117         print("Deletion confirmed via status code.")
118     else:
119         print("Deletion failed based on status code.")
120
121     print("\nStudent API tests complete.")
122
123
124 if __name__ == "__main__":
125     main()

```

Output:

```

{"endpoints":{"DELETE /students/<id>":"Delete a student","GET /students":"List all students","POST /students":"Create a student (name:str, age:int, major:str)","PUT /students/<id>":"Update a student"},"message":"Student API is running"}

```

Observation:

Server is running locally at <http://127.0.0.1:5000> and responds reliably.

Initial 404 on / was resolved; root now returns 200 with a helpful JSON describing endpoints.

CRUD flow behaves correctly and consistently across multiple cycles:

GET /students: 200 with list (empty after fresh start).

POST /students: 201 with created student and incremental id.

PUT /students/<id>: 200 with updated fields.

DELETE /students/<id>: 204 with no body (expected for successful deletion).

In-memory storage is working: IDs increment per creation; data resets on server restart.

Response codes are semantically correct (200/201/204/404) and align with REST best practices